task1.py

```
1    print("Hello Hunarmand Punjab")
```

TERMINAL

```
Hello Hunarmand Punjab
```

task2.py

```
1    string = "This is variable"
2    integer = 45
3    floting = 54.6
4    isTrue = True
```

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task3.py

```python
a= 5
b = 6

print(a + b )
print(a - b)
print(a*b)
print(a/b)
print(a%b)

```

TERMINAL

```
11
-1
30
0.8333333333333334
5
```

task4.py

```
1    a = 45
2    b = 56.56
3    print(int(b))
4    print(float(a))
```

TERMINAL

```
56
45.0
```

task1.py

```python
a = 5
b = 4
# Arithmetic operators: +, -, *, /, %, //, **
print(a+b) # Add a and b
print(a-b) # subtract b form a
print(a*b) # multiply a with b
print(a/b) # a divide b
print(a%b) # modulus (reminder)
print(a//b)# Floor division
print(a**b)# a square b
#  Comparison operators: ==, !=, <, >, <= , >=

print(a==b) # Is a equal b (True or False)
print(a!=b) # Is not equal b (True or False)
print(a<b)  # Is a les than b (True or False)
print(a>b)  # is a greater than b (True or False)
print(a<=b) # is a less than or equal to b (True or False)
print(a>=b) # is a greater than or equal to b (True or False)

# Logical operators: and, or, not

bool1 = True
bool2 = False

print(bool1 and bool2) # is both are True == True
print(bool1 or bool2)  # is one of these area True == True
print(not bool2) # return  oposit of bool2

# Assinment operators
a= 4
a+= 5
a-=6
a *=5


# bitwise oprator
print(a & b)
print( a | b)

```

TERMINAL

```
9
1
20
1.25
1
1
625
False
True
False
True
False
True
False
True
True
4
15
```

task2.py

```
1    a = 4 + 5 >6 ==5
2    print(a)
```

TERMINAL

```
False
```

task1.py

```
1    num = int(input("Enter Number : "))
2    if num&2==0:
3        print("Number in Even")
4    elif num == 0 :
5        print("number is zero")
6    else:
7        print("number is odd")
```

TERMINAL

```
[Execution timed out]
```

task2.py

```python
marks = int(input("Enter marks: "))
if marks<50:
    print("fail")
elif marks<60:
    print("D")
elif marks<70:
    print("C")
elif marks<80:
    print("B")
elif marks<90:
    print("A")
elif marks<100:
    print("A+")
```

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```
[Execution timed out]
```

task1.py

```python
for i in range(21):
    print(i)
```

TERMINAL

```
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
```

task2.py

```python
i = 2
while i<20:
    if i % 2 ==0:
        print(i)
    i+=1
```

TERMINAL

```
2
4
6
8
10
12
14
16
18
```

task3.py

```python
for i in range(8,10):
    for j in range(1,11):
        print('{0} * {1} = {2}'.format(i, j, i*j))
    print("-----------")
```

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```
8 * 1 = 8
8 * 2 = 16
8 * 3 = 24
8 * 4 = 32
8 * 5 = 40
8 * 6 = 48
8 * 7 = 56
8 * 8 = 64
8 * 9 = 72
8 * 10 = 80
-----------
9 * 1 = 9
9 * 2 = 18
9 * 3 = 27
9 * 4 = 36
9 * 5 = 45
9 * 6 = 54
9 * 7 = 63
9 * 8 = 72
9 * 9 = 81
9 * 10 = 90
-----------
```

task1.py

```
1    def add(a,b):
2        return a+b
3
4    print(add(5,9))
```

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```
14
```

task2.py

```python
def isPrime(num):
    i = 2
    while i<num:
        if num%i == 0:
            return False
        i+=1

    return True

n = int(input("Enter number: "))
print(isPrime(n))
```

TERMINAL

```
[Execution timed out]
```

task3.py

```python
# Defalt parameters
def greet(name="Guest"):
    print(f"Hello, {name}!")

greet()
greet("Ali")

# *args
def add_numbers(*args):
    total = sum(args)
    return total

print(add_numbers(1, 2, 3))
print(add_numbers(10, 20, 30, 40))

# **kwargs
def display_info(**kwargs):
    for key, value in kwargs.items():
        print(f"{key}: {value}")

display_info(name="Ali", age=25, city="Lahore")
```

TERMINAL

```
Hello, Guest!
Hello, Ali!
6
100
name: Ali
age: 25
city: Lahore
```

task1.py

```
1    colors = ["red", "Green", "Yellow", "Blue"]
2    for color in colors:
3        print(color)
```

TERMINAL

```
red
Green
Yellow
Blue
```

task2.py

```python
colors = ("red","green","yellow")

colors[0]= "blue"  # this will be show error
```

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```
Traceback (most recent call last):
  File "d:\VScode\LMS\ass 2\part 6\task2.py", line 3, in <module>
    colors[0]= "blue"  # this will be show error
    ~~~~~~~^^^
TypeError: 'tuple' object does not support item assignment
```

task3.py

```python
set1 = {1,2,3,4,5,6}
set2 = {10,11,12,13,14,15}

# Union
print("Union:", set1.union(set2))

# intersection
print("Intersection:", set1.intersection(set2))

# difference
print("Difference:", set1.difference(set2))


```

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```
Union: {1, 2, 3, 4, 5, 6, 10, 11, 12, 13, 14, 15}
Intersection: set()
Difference: {1, 2, 3, 4, 5, 6}
```

tasks.py

```python
student = {"name":"Noman",
           "age":18,
           "Grade":"A"}

# access and update values
student["name"]= "Noman Tahir"
student["Grade"]= "A+"
student["age"] = 19
# print(student)

# Loop through dictionary items
for key , value in student.items():
    print(f"{key} , {value}")
```

TERMINAL

```
name , Noman Tahir
age , 19
Grade , A+
```

tasks.py

```python
with open("sample.txt","w") as file:
    file.write("Hellow Noman")

#  Read
with open("sample.txt","r") as fileRead:
    show = fileRead.read()

print(show)

#  Append
with open("sample.txt","a") as fileAppend :
    fileAppend.write("this line is appened")
```

TERMINAL

```
Hellow Noman
```

tasks.py

```python
import math as mt
import random as ram
print(mt.sqrt(25))
print(mt.factorial (25))
print(ram.randint(25,30))

```

TERMINAL

```
5.0
15511210043330985984000000
25
```

tasks.py

```python
a = 10
try:
    print(a/0)
except ZeroDivisionError:
    print("Any number cant be divided to zero")


try:
    print(a/0)
except ZeroDivisionError:
    print("Any number cant be divided to zero")
except ValueError:
    print("Value is ")
```

TERMINAL

```
Any number cant be divided to zero
Any number cant be divided to zero
```

tasks.py

```python
class Student:
    def __init__(self, name, age, grade):
        self.name = name
        self.age = age
        self.grade = grade

    def display_info(self):
        print(f"Name: {self.name}, Age: {self.age}, Grade: {self.grade}")

class Teacher(Student):
    def __init__(self, name, age, grade, subject):
        super().__init__(name, age, grade)
        self.subject = subject

    def display_info(self):
        super().display_info()
        print(f"Subject: {self.subject}")
```

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task.py

```python
class Student:
    def __init__(self, name, age, grade):
        self.name = name
        self.age = age
        self.grade = grade


students = []

def load_data():
    try:
        with open("students.txt", "r") as file:
            for line in file:
                data = line.strip().split(",")
                if len(data) == 3:
                    student = Student(data[0], data[1], data[2])
                    students.append(student)
    except FileNotFoundError:
        pass

def save_data():
    with open("students.txt", "w") as file:
        for student in students:
            file.write(f"{student.name},{student.age},{student.grade}\n")

def add_student():
    name = input("Enter name: ")
    age = input("Enter age: ")
    grade = input("Enter grade: ")
    student = Student(name, age, grade)
    students.append(student)
    save_data()
    print("Student added successfully!")

def view_students():
    if len(students) == 0:
        print("No students found.")
    else:
        print("\n--- All Students ---")
        for i in range(len(students)):
            s = students[i]
            print(f"{i + 1}. Name: {s.name}, Age: {s.age}, Grade: {s.grade}")
        print("--------------------")

def update_student():
    view_students()

    if len(students) > 0:
        index = int(input("Enter student number to update: ")) - 1

        if 0 <= index < len(students):
            s = students[index]
            s.name = input("Enter new name: ")
            s.age = input("Enter new age: ")
            s.grade = input("Enter new grade: ")

            save_data()
            print("Student updated successfully!")
        else:
            print("Invalid number.")

def delete_student():
    view_students()
    if len(students) > 0:
        index = int(input("Enter student number to delete: ")) - 1

        if 0 <= index < len(students):
            students.pop(index)

            save_data()
            print("Student deleted successfully!")
        else:
            print("Invalid number.")

def main():
    load_data()

    while True:
        print("\nMenu:")
        print("1. Add student")
        print("2. View all students")
        print("3. Update student info")
        print("4. Delete student")
        print("5. Exit")

        choice = input("Enter choice (1-5): ")

        if choice == "1":
            add_student()
        elif choice == "2":
            view_students()
        elif choice == "3":
            update_student()
        elif choice == "4":
            delete_student()
        elif choice == "5":
            print("Goodbye!")
            break
        else:
            print("Invalid choice. Please try again.")

main()
```

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```
[Execution timed out]
```